MILITARY
SPECIFICATION

MIL-M-13787(SigC)
16 November 1954

MOTORS, ALTERNATING CURRENT, FRACTIONAL HORSEPOWER, SQUIRREL CAGE (FOR COMMUNICATION AND OTHER ELECTRONIC AND SPECIAL MILITARY APPLICATIONS)

1. SCOPE

1.1 This specification covers alternating current, fractional horsepower motors, having the specific environmental characteristics required for communication and other electronic and special military applications, and capable of withstanding extreme service and environmental conditions such as shock, high temperature, low temperature, vibration, humidity, fungus etc., A fractional horsepower motor is a motor built in a frame smaller than that of a motor having a continuous rating of one horsepower open construction, at 1,700 to 1,800 revolutions per minute (i. e., the outside diameter of the frame shall be less than 10 inches.) This specification covers all types of squirrel-cage induction motors but does not cover control motors, servomotors, or any motor types which incorporate brushes. This general specification is applicable as a procurement document when used with a complementary document (. e. g., specification sheet, service drawing, or other document) giving detail requirements and applicable additions or exceptions to the general requirements and tests specified herein.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

FF-B-171	Bearings, Ball
NN-B-621	Boxes; Wood, Nailed and Lock Corner
NN-B-631	Boxes; Wood, Wirebound (For Domestic Shipment)
QQ-S-571	Solder, Soft (Tin, Tin-Lead, and Lead-Silver)
UU-T-111	Taper, Paper, Gummed (Kraft)
LLL-F-291	Fiberboard; Corrugated, Single-Face (Flexible)
LLL-B-631	Boxes; Fiber, Corrugated (For Domestic Shipment)
PPP-B-601	Boxes; Wood-Cleated-Plywood

MILITARY

MIL-P-14	Plastic-Materials, Molding, and Plastic-Parts, Molded; Thermosetting
MIL-P-79	Plastic-Materials, Laminated, Thermosetting Rods and Tubes
JAN-P-106	Packaging and Packing for Overseas Shipment- Boxes, Wood Nailed, (For Weight of Contents Not in Excess of 1000 Pounds)
JAN-P-108	Packaging and Packing for Overseas Shipment- Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior
MIL-P-116	Preservation, Methods Of

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JAN-B-121	Barrier Materials, Greaseproof
MIL-C-124	Containers (Cans, Pails and Drums), Metal (For other Than Subsistance Items)
JAN-P-127	Packaging and Packing for Overseas Shipment-Tape, Adhesive, Pressure-Sensitive, Water-Resistant
MIL-B-131	Barrier Material, Water-Vapor-Proof, Flexible
JAN-P-133	Packaging and Packing for Overseas Shipment-Boxes, Set-Up, Paperboard
JAN-T-152	Treatment, Moisture and Fungus Resistant, of Communications, Electronic, and Associated Electrical Equipment, General Process For
JAN-E-251	Electrical-Equipment; Rotation, Connections, and Terminal Markings For
JAN-W-583	Wire, Magnet
MIL-S-901	Shockproof Equipment, Class HI- (High-Impact) Shipload.
MIL-P-997	Plastic-Material, Laminated, Thermosetting, Electrical-Insulating; Sheets, Glass Cloth, Silicone Resin
MIL-V-1137	Varnish, Electrical-Insulating (For Electro-Motive Equipment)
MIL-P-3115	Plastic-Material, Laminated, Thermosetting, Sheets, Paper-Base, Phenoli c-Resin
MIL-G-3278	Grease; Aircraft and Instruments (For Low and High Temperature)
MIL-E-5272	Environmental, Testing, Aernautical and Associated Equipment, General Specification For
MIL-O-5606	Oil; Hydraulic, Aircraft, Petroleum Base
MIL-L-6085	Lubricating Oil; Aircraft Instrument, Low Volatility
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys and Nickel Alloys
MIL-T-10513	Tropicalization of Signal Corps Equipment
MIL-M-13231	Marking of Electronic Items
MIL-P-15037	Plastic-Material, Laminated Thermosetting, Sheets, Glass-Cloth Melamine-Resin
MIL-P-15047	Plastic-Material, Laminated Thermosetting, Sheets, Nylon Fabric Base, Phenolic -Resin

U. S. ARMY

19-10	Filler, Interior Packing, Fiberboard, Corrugated, Single and Double-Faced, For General Packing Purposes
72-53	Finishes (For Ground Signal Equipment)

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U. S. AIR FORCE

7201	Test Procedure: Shock, Medium Impact, Variable-Duration, Method and Apparatus For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment
MIL-STD-202	Test Methods For Electronic and Electric Components Parts

SPECIFICATION SHEETS

(Specification sheets for motors covered by this specification, when issued, will be in supplement to this specification)

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Service

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Preproduction samples.-

3.1.1 Quantities.- After award of contract, six samples of each type of motor on order shall be furnished for Government inspection and approval.

3.1.2 Compliance with specified requirements.-

3.1.2.1 Preproduction samples shall meet tspecified requirements, shall be made and assembled by tools and methods that will be used for quantity production and shall be accompanied by a statement to that effect. (See 3.1.2.3 for exceptions).

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3.1.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (See 3.1.5), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.1.2.3 When devision from 3.1.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and the manner inwhich it will be corrected in production of motors on contract.

3.1.3 Test data. - The preproduction samples shall be accompanied by test data showing compliance with specific performance. The test data shall comprise an engineering report giving test procedures, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like).

3.1.4 Preproduction inspection. - Preproduction samples will be subjected by the Government to the preproduction inspection tests specified in Table I, in the order shown. After completion of Group 1 tests, the samples will be divided into two equal groups and subjected to the tests for the particular group.

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Table I - Preproduction inspection

Test	Requirement paragraph	Test Paragraph
Test Group I (all of sample)		
Visual and Mechanical Inspection 1/	3.9 to 3.9.2.2 incl. 3.9.3 3.9.4 3.23.4 to 3.23.8 incl. 3.23.11 3.24	4.2.1
Terminal or lead strength	3.10	4.2.2
Insulation resistance	3.11	4.2.3
Dielectric strength	3.12	4.2.4
Concentricity	3.13	4.2.5
Dynamic balance	3.14	4.2.6
Speed-torque characteristic 2/	3.26	4.2.7
Temperature rise 3/	3.26	4.2.8
Temperature cycling	3.15	4.2.9
Test Group II (1/2 of sample)		
Mechanical shock	3.18	4.2.11
Voltage and frequency variation	3.19	4.2.12
Low-temperature storage and operation (when applicable)	3.20	4.2.14
Moisture resistance	3.21	4.2.15
Tropicalization	3.6	4.2.19
Visual and mechanical inspection 1/	3.2.1 3.2.2 3.2.4 3.2.5 to 3.9.2.2 incl. 3.24 3.25 3.26	4.2.1
Test Group III (1/2 of sample) 4/		
Vibration	3.16	4.2.10
Life	3.26	4.2.16
Corrosion	3.22	4.2.17
Motor enclosure	3.23.1 to 3.23.3 incl	4.2.18
Visual and mechanical inspection 1/	3.2.1 3.2.2 3.2.4 3.2.5 to 3.9.2.2 incl 3.26 3.24 3.28	4.2.1

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1/ Marking will be considered defective only if it is illegible at the completion of any of the required tests.
2/ Only three specimens will be subjected to the speed-torque tests.
3/ Only two specimens will be subjected to the temperature-rise test.
4/ One of the specimens which has been subjected to the temperature-rise test will be included in each of test groups II and III.

3.1.5 Reference standards. - After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.2 Material. -

3.2.1 Restricted materials. -

3.2.1.1 Inflammable and explosive materials. - Motors shall incorporate in their construction the smallest practicable amount of inflammable and explosive materials consistant with good commerical practice.

3.2.1.2 Corrosive materials. - Corrosive materials used in any of the processes in the manufacture of motors shall be neutralized to minimize corrosion in the completed unit. Materials used in the construction of motors shall be of a corrosion-resistant type of shall be treated to resist corrosion.

3.2.2 Plastic. -

3.2.2.1 Laminated. - Laminated plastic sheets shall be in accordance with Specification MIL-P-996, MIL-P-3115, MIL-P-15037 or MIL-P-15047, as applicable. Laminated plastic rods and tubes shall be in accordance with Specification MIL-P-79. All tubes shall be seamless rolled.

3.2.2.2 Molded. - Molded plastic-material except where used for wearing parts of switch mechanisms, shall be in accordance with Specification MIL-P-14. Where a high degree of arc resistance is required, type CMG, MME, or CMI-5 shall be used.

"3.2.3. - Electrical Insulation. - Motor windings which operate at a hot-spot temperature (maximum ambient temperature plus temperature rise) of 105°C or less shall contain Class A insulation. Motor windings which operate at a hot-spot temperature between 105°C and 135°C shall contain Class B insulation. Motor windings which operate at a hot-spot temperature between 135°C and 200°C shall contain Class H insulation (see 6.8)"

3.2.4 Wire. - Magnet wire shall conform to Specification JAN-W-583; however, wire having a diameter smaller than 0.0031 inch (AWG size 40) and wire used in motors having Class B or H insulation (See 6.8) shall require prior approval by the contracting officer. (See 3.2.3) Lead wire shall be insulated, flexible, stranded wire.

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3.2.5 Lubricants. - Lubricants supplied in motors which require relubrication shall conform to Specification MIL-G-3278 or MIL-L-6085, as applicable.

3.3 Design and construction. -

3.3.1 Bearings. - Ball bearings, when used, shall conform to Specification FF-B-171.

3.3.1.1 Bearing mounting. - When bearings are mounted in soft metal such as zinc alloys, aluminum, or copper, a steel insert shall be provided when specified.

3.3.2 Impregnation. - Windings shall be suitably impregnated. Impregnating materials and processes shall be such that there will be no flowing, cracking, chipping, crazing, or giving off of fumes that will intefere with noraml motor operation. Impregnation varnish shall conform to Specification MIL-V-1137, Grade CB, unless otherwise specified by the contracting officer.

3.4 Brazing and soldering. -

3.4.1 Brazing. - Brazing of steel, copper, copper alloys, and nickel alloys shall conform to Specification MIL-B-7883.

3.4.2 Soldering. -

3.4.2.1 Solder. - Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ-S-571. Solder shall not be used primarily for obtaining mechanical strength. All external terminals used for solder connections shall be hot-tin-dipped to facilitate soldering.

3.4.2.2 Flux and cleaning agents. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.4.2.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. The soldering shall be so accomplished that no rosin remains on the joint or in the proximity of the joint. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

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parts, after fabrication, shall be cleaned in accordance with goe, or as specified in an applicable document. Cleaning pro - ceterious effect on the equipment. Corrosive material shall be rere parts are assembled into the equipment. After assembly, coned thoroughly and shall be free from particles of solder flux, aml. If necessary, such cleaning shall also be performed before finents.

The assembled equipment shall be treated in accordance with Sp Before being assembled into the equipment, materials and pai accordance with Specification MIL-T-10 513.

shall be of ample size. Where tack or spot welds are employed, thc be sufficient to provide adequate strength and at le ast two wdd each welded part. Spot or tack welds in aluminum or mag- neo lightly stressed, noncritical structures. Wherever possible, thesuch that grinding on the finished weld will be unnecessary.

:faces to be welded shall be cleaned in accordance with good co: shall be free from rust, scale, paint, grease, and other forl assemblies shall be cleaned to remove scale, oxidation prc by sandblasting, wire brushing, or other suitable means. Acall be completely neutralized and removed.

heating shall be employed where distortion is likely to result frcll have thorough penetration and good fusion and shall be free frcormal pock marks, and other harmful defects. Where unde- sirare likely to result from welding, the welding assembly shall bet-gas -shielded arc welding shall be used, when practicable, fopr magnesium.

eads. - Stator leads shall be firmly secured to the stator wineans in addition to the solder connection. Where leads are brc means shall be provided so that no external strain can be trg, unless otherwise specified.

ıreaded parts, and related items. -

- Screw threads shall conform to Handbook H28 and supplement theead Form and Series shall be class 2A or 2B fit and the And Series shall be class 2 fit. Except for holding labels or sirnot used for structural purposes or critical applications, thither soft material shall not be employed where other means of fasand, in particular, shall not be used when frequent assembly anecessary. When specified, steel inserts shall be provided for thc metal which will be detached from the motor in normal opera- tioen considered necessary by the Government inspector, based onquipment or observation of the contractor's production methods, thc torque measuring or limiting device for tightening threaded pas are used, the values of torque shown in table II shall be apWhen torque measuring or limiting devices are used for screws

Sheet #6

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of other size, pitch, or material, the torque shall be the design value established by the contractor for the particular application.

TABLE II - Torque for tightening steel screws

Thread	Torque (in-lb)
6-32	9 to 12
8-32	18 to 24
10-24	29 to 34
10-32	32 to 40
12-24	42 to 56

3.9.2 Thread engagement and extension. - The length of thread engagement for tapped holes in steel or hard brass shall at least equal the nominal thickness of corresponding machine-screw nuts in sizes No. 12 and smaller and semifinished hexagon regular nuts in sizes 1/4 inch and larger, as shown in Handbook H28. Thread engagement for tapped holes in aluminum, magnesium, or other soft material shall be at least 50 percent greater than the thread engagement specified for steel or hard brass. Where permitted by design of the equipment, screws and bolts shall extend at least 1-1/2 threads beyond the nut.

3.9.2.1 Locking of screw-thread assemblies. - All screw-threads assemblies shall be rendered resistant to loosening under vibration.

3.9.2.2 Through-bolting. - Through-bolting shall be used wherever practicable.

3.9.3 Finish. - The finish shall be in accordance with Specification 72-53, except that crackle or wrinkle finish shall not be used.

3.9.4 Direction of rotation. - Facing the end of the motor opposite the drive shaft extension, the direction of rotation shall be determined as being either clockwise, counterclockwise, or reversible, as specified.

3.9.5 Run-in period. - All motors shall be subjected to a run-in period.

3.9.6 Duty cycle. - Motors shall be designed to operate at one of the following duty cycles, as specified.

3.9.6.1 Continuous. - A continuous motor is one designed for operation at full-load torque continuously at the specified maximum ambient temperature without exceeding the specified temperature rise.

3.9.6.2 Intermittent. - An in termittent motor is one designed for operation at specified regular intervals at a specified combination of full-load torque, no load, and rest.

3.10 Terminal or lead strength. - The terminals or lead shall show no damage when tested as specified in 4.2.2.

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3.11 Insulation resistance. - The motor shall have an insulation resistance of not less than 12 megohms, when tested as specified in 4.2.3.

3.12 Dielectric strength. - There shall be no puncture of solid insulation, no continuing sparkover, and no breakdown in dielectric strength and no damage to the insulation sufficient to cause failure of the motor, when tested as specified in 4.2.4.

3.13 Concentricity. - When motors are tested as specified in 4.2.5, there shall be no rubbing or binding of the rotor.

3.14 Dynamic balance. - When motors are tested as specified in 4.2.6, the maximum total excursion shall not exceed 0.0015 inch unless otherwise specified.

3.15 Temperature cycling. - When tested as specified in 4.2.9, the motor shall meet the requirements for dielectric strength, (See 3.12) and there shall be no impairment of protective coatings.

3.16 Vibration. - When motors are tested as specified in 4.2.10, there shall be no mechanical or electrical damage, and the motor shall be capable of operation.

3.17 Overload. - When motors are tested as specified in 4.2.13, the no load speed shall not change by more than 10 percent from the value previously determined in the no-load test (See 4.2.7.4) and the motor shall meet the requirements for dielectric strength (See 3.12)

3.18 Mechanical shock. -

3.18.1 Test 1. - The motor shall be operative after each blow, and there shall be no binding or rubbing of parts, when tested as specified in 4.2.11.1. Momentary stopping is permissable provided the motor resumes operations after each blow.

3.18.2 Test 2. - There shall be no mechanical or electrical damage, when the motor is tested as specified in 4.2.11.2.

3.19 Voltage and frequency variation. - The motor shall start and operate under the conditions specified in 4.2.12.

3.20 Low-temperature storage and operation (when applicable). - The motor shall reach 90 percent of specified full-load speed within 30 seconds for continuous motors, and within 5 seconds for intermittent motors and shall continue to operate for the specified period, when tested as specified in 4.2.14.

3.21. Moisture resistance. - After the test specified in 4.2.15, the insulation resistance of the motor shall be not less than 50,000 ohms when tested within one hour after removal from the chamber. The motor shall meet the requirements for dielectric strength (See 3.12), the full-load speed shall be as specified and the marking shall remain legible. After the 48-hour drying period, the insulation resistance shall be not less than 12 megohms.

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3.22 Corrosion. - There shall be no evidence of excessive corrosion, and the marking shall remain legible, whe n motors are tested as specified in 4.2.17.

3.23 Motor enclosure. - The motor enclodure shall conform to one or more of the following enclosures.

3.23.1 Explosionproof. - When tested as specified in 4.2.18.1, the enclosure shall not permit the ignition of the explosive vapor surrounding the motor in the test chamber by transmission of the internal explosion.

3.23.2 Watertight. - When tested as specified in 4.2.18,4, the motor shall show no evidence of water penetration, except that leakage may occur around the shaft, provided water is prevented from entering the oil reservoir and provision is made for draining the motor.

3.23.3 Submersible. - When tested as specified in 4.2.18.5, the motor shall show no evidence of water penetration.

3.23.4 Open. - An open motor shall have ventilating openings which permit passage of external cooling air over and around the windings.

3.23.5 Protected. - A protected motor shall have ventilating openings which shall not exceed 1/2 square inch in area and shall be of such shape as to prevent the passage of a rod 1/2 inch in diameter.

3.23.6 Totally enclosed. - A totally enclosed motor shall be so enclosed as to prevent the free exchange of air between the inside and the outside of the case, but not sufficiently enclosed to be termed airtight.

3.23.7 Fan-cooled. - A fan-cooled motor shall incorporate an external fan mounted on the shaft as an aid in cooling. Drain holes shall be provided for the rapid drainage of water from the outer or fan enclosure.

3.23.8 Dripproof. - A dripproof motor is an open motor in which the ventilating openings are so constructed that drops of liquid or solid particles falling on the motor at any angle not greater than 15° from the vertical cannot enter the motor, either directly or by striking and running along a horizontal or inwardly inclined surface.

3.23.9 Oilproof. - When tested as specified in 4.2.18.3, the motor shall be operative and shall meet the requirements for dielectric strength.

3.23.10 Dustproof. - When tested as specified in 4.2.18.2, the motor shall show no evidence of water penetration.

3.23.11 Splashproof. - A splashproof motor is an open motor in which the ventilating openings are so constructed that drops of liquid or solid particles falling on the motor or coming toward it in a straight line at any angle not greater than 100° downward from the vertical cannot enter the motor either directly or by striking and running along a surface.

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3.24 Marking.- The motor shall be permanently and legibly marked with the type of motor (e. g. , split phase), the specification sheet number, the manufacturer's name or symbol, and any other information specified. Terminal marking shall conform to Specification JAN-E-251.

3.25 Connections.- The electrical connections of motors shall not depend on wires, lugs, terminals, or other connectors clamped between a metallic member and an insulating material. When the maintenance of a tight connection depends on the resistance of an insulating material to compressive or shearing stress, such a connection shall be securely soldered. Electrical connections shall be made mechanically secure and electrically continuous before soldering.

3.26 Complementary documents for individual motor types.- The following requirements, if specified on complementary documents, (e. g. service drawings, specifications or other documents) shall be tested for in accordance with the corresponding test paragraphs as indicated in Section 4 of this specification.

1 - Free end play and clearance	(See 4. 2. 1)
2 - Dimensions and weight	(See 4. 2. 1)
3 - Minimum locked-rotor torque and minimum locked-rotor current	(See 4. 2. 7. 1)
4 - Full load speed, full-load current, full-load power input	(See 4. 2. 7. 2)
5 - Efficiency	(See 4. 2. 7. 3)
6 - No load	(See 4. 2. 7. 4)
7 - Breakdown torque	(See 4. 2. 7. 5)
8 - Pull-up torque, switching torque and switching speed	(See 4. 2. 7. 6)
9 - Temperature rise of winding and frame	(See 4. 2. 8)
10 - Life	(See 4. 2. 16)

3.26.1 When requested, the manufacturer shall furnish the Government complete information (mfg. drawings, special processes, types and makes of all material, test data) covering the manufacture and testing of equipment procured under this speciffication.

3.26.2 In the event of a conflict between the requirements of this general specification and the complementary document, the latter shall govern.

3.27 Serial numbers.- Serial numbers under Specification MIL-M-13231 are not required.

3.28 Workmanship.- The equipment shall be manufactured and assembled in a thoroughly workmanlike manner in accordance with the applicable portions of paragraphs 3. 2. 1 through 3. 9. 3, and all components, subassemblies, parts and final assembly shall be examined for defects of quality as listed in 4. 2. 1.

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4. QUALITY, ASSURANCE PROVISIONS

4.1 Inspection, general.- Government inspection shall consist of group A, group B, and group C inspection as specified in 4.1.1 through 4.1.8. (See 6.10.1). Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B or group C inspection, or both.

4.1.1 Group A inspection.- Group A inspection shall be in accordance with table III and Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

4.1.2 Standard test conditions.- Unless otherwise specified herein, all measurements and tests shall be made at 25 $^{+10}_{-5}$ C., ambient atmospheric pressure, and ambient humidity.

4.1.3 Test equipment and inspection facilities. - Test equipment and inspection facilities shall be of sufficient accuracy and quality to permit performance of the required inspection tests. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government.

4.1.4 Group A tests.- Group A tests shall consist of those specified in table III in the order shown. Group A tests shall be performed by either of the following methods or a combination of both.

(a) 100-percent inspection

(b) Statistical sampling and inspection in accordance with Standard MIL-STD-105. The acceptable quality levels (AQL) for visual and mechanical inspection (external) shall be 1.5 percent defective) for major defects and 6.5 (percent defective) for minor defects. Major and minor defects shall be as defined in Standard MIL-STD-105. For all other group A tests combined, the AQL shall be 1.5 (percent defective) for major defects; all defects shall be considered major.

4.1.4.1 Lot.- An inspection lot shall be as specified in Standard MIL-STD-105.

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Table III - Government Group A inspection

Test	Requirement paragraph	Test paragraph
Visual and mechanical inspection (External)		4.2.1
Threaded parts	3.9 to 3.9.2.2 incl.	
Finish	3.9.3	
Marking	3.24	
Workmanship	3.28	
Insulation Resistance	3.11	4.2.3
Dielectric Strength	3.12	4.2.4
Concentricity	3.13	4.2.5
Dynamic balance	3.14	4.2.6
Speed-torque characteristics		
Locked rotor	3.26	4.2.7.1
Full load	3.26	4.2.7.2
No load	3.26	4.2.7.4
Breakdown	3.26	4.2.7.5

4.1.5 Group B inspection. - Group B inspection shall be in accordance with table IV.

4.1.5.1 Order of inspection within Group B. - Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.1.5.2 Sampling procedure. - Sample Units shall be selected, without regard to their quality, by the Government inspector. Unless otherwise specified, normal inspection shall be used at the start of the contract. If the number of defective sample units from a lot exceeds the acceptance number specified in the sampling plan, the lot shall be rejected. Disposition of the rejected product (sample units and lots) shall be in accordance with Standard MIL-STD-105 and 4.1.10.

4.1.5.3 Procedure in case of failure. - When a lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

Table IV - Government Group B inspection

Test	Rqt. Par.	Test Par.	Group B Plans	Table
Terminal or lead strength	3.10	4.2.2	B-2	V
Voltage and frequency variation	3.19	4.2.12	B-2	V
Low-temperature storage and Operation (when applicable)	3.20	4.2.14	B-2	V

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Table Va .- Group B-2 sampling plans

Lot size /&	Reduced /# Inspection		Normal /# Inspection		Tightened /# Inspection	
	Sample size	Acceptance number	Sample size	Acceptance number	Sample size	Acceptance number
9 to 25, incl.	2	0	2	0	4	0
26 to 65, incl.	2	0	3	0	6	0
66 to 180, incl.	7	1	7	1	17	1
181 to 500, incl.	7	1	17	2	28	2
501 to 1,300, incl.	7	1	28	3	41	3
1,301 to 3,200, incl.	17	2	41	4	53	4
3,201 to 8,000, incl.	17	2	53	5	66	5
8,001 to 22,000, incl.	28	3	66	6	81	6
22,001 and over	28	3	81	7	81	6

/& For lot sizes under 9, the sample size shall be at the option of the Government but not more than 2, and the acceptance number shall be 0.

/# Normal, tightened, or reduced inspection shall be used in accordance with the following: (Part "b" of this table lists values for limits A, B, and C, referenced below.)

(1) Calculate the estimated process average from the formula 100 P/N; where P equals the number of defective units found in N sample units, and N equals the number of sample units included in the estimated process average. The Government will determine the number of consecutive sample units to be used to estimate the process average.

(2) Except as stated in (3) below, use normal inspection as long as the process average is between limits A and C.

(3) Change to tightened inspection whenever the process average becomes equal to or greater than limit A, or whenever a lot fails to pass.

(4) Return to normal inspection from tightened inspection when two lots in succession pass tightened inspection, and the process average becomes less than limit B.

(5) Change to reduced inspection whenever the process average becomes less than limit C.

(6) Return to normal inspection from reduced inspection when the process average becomes equal to or greater than limit C.

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Table Vb .- Group B-2 Limits of the process average

N	Limit A %	Limit B %	Limit C %	N	Limit A %	Limit B %	Limit C %
10	18.78	11.89	*	130-144	8.82	6.91	1.18
11	18.14	11.57	*	145-159	8.62	6.81	1.38
12	17.58	11.29	*	160-179	8.45	6.72	1.55
13	17.09	11.04	*	180-199	8.25	6.62	1.75
14	16.65	10.82	*	200-249	8.08	6.54	1.92
15-16	16.25	10.63	*	250-299	7.76	6.38	2.24
17-18	15.57	10.29	*	300-349	7.52	6.26	2.48
19-20	15.00	10.00	*	350-399	7.33	6.16	2.67
21-23	14.51	9.76	*	400-449	7.18	6.09	2.82
24-26	13.90	9.45	*	450-499	7.05	6.03	2.95
27-29	13.39	9.19	*	500-549	6.95	5.97	3.05
30-34	12.96	8.98	*	550-599	6.86	5.93	3.14
35-39	12.37	8.68	*	600-649	6.78	5.90	3.22
40-44	11.89	8.45	*	650-699	6.71	5.85	3.29
45-49	11.50	8.25	*	700-749	6.65	5.82	3.35
50-59	11.16	8.08	*	750-799	6.59	5.80	3.41
60-69	10.63	7.81	*	800-849	6.54	5.77	3.46
70-79	10.21	7.60	*	850-899	6.50	5.75	3.50
80-89	9.87	7.44	0.13	900-949	6.45	5.73	3.55
90-99	9.59	7.30	0.41	950-999	6.41	5.71	3.59
100-114	9.36	7.18	0.64	1000	6.38	5.69	3.62
115-129	9.06	7.03	0.94				

* Number of sample units included in estimated process average is insufficient for reduced inspection.

N = Number of sample units in estimated process average.

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4.1.6 Group C inspection. - This shall consist of the group C inspection specified in Table VI.

4.1.7 Selection of samples. - One motor shall be selected from the first lot presented for inspection which has been subjected to and has passed the group A tests but which has not been subjected to the group B tests. When production continues for a period beyond the rated life of the sample unit under test, an additional sample unit selected from current production shall be placed on test periodically at intervals equal to the rated life of the motor, except that these intervals shall be not less than 30 days.

4.1.8 Noncompliance. - If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of corrective action taken on the process and affected portion of product. If the Government inspector does not consider that the corrective action will enable the equipment to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

Table VI - Group C inspection tests.

Test	Requirement Paragraph	Test Paragraph
Temperature rise	3.26	4.2.8
Life	3.26	4.2.16
Visual and mechanical inspection (internal)		4.2.1
Restricted materials	3.2.1	4.2.1
Plastic	3.2.2	4.2.1
Wire	3.2.4	4.2.1
Lubricants	3.2.5	4.2.1
Bearings	3.3.1 and 3.3.1.1	4.2.1
Impregnation	3.3.2	4.2.1
Securing of wire leads	3.8	4.2.1
Threaded parts	3.9 to 3.9.2.2 incl	4.2.1
Free end-play and clearance	3.26	4.2.1
Dimensions and weight	3.26	4.2.1

4.1.9 Reinspection of conforming group B and group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are re-subjected to and pass group A inspection (after repair of all visible damage).

4.1.10 Disposition of nonconforming product. - When defective sample units or rejected lots are resubmitted or affected portion of product is submitted for acceptance, such items shall be suitably tagged or identified by equivalent means to indicate the

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cause of failure and means employed to correct the fault. The record shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.2 Test procedures. -

4.2.1 Visual and mechanical inspection. - Parts and equipment shall be examined for defects, classification of defects should include those listed below as major and minor.

1 - Major
- a - Loose nuts and screws
- b - Welding-scabs, blisters, abnormal pock marks and other harmful defects.
- c - Soldering-less than 1/2 turn, insufficient solder, movement of wire or solder upon pick application, insulation in terminal hole.
- d - Wiring-broken strands, damaged or frayed insulation, wire exhibits no slack.
- e - Riveting-rivets too close to the edge of the material riveted, damage to the joined parts, rivet head improperly seated.
- f - Painting-Color not as specified, unsatisfactory adhesion, film thickness less than 1/2 specified thickness.
- g - Marking-Missing, illegible or wrong marking.
- h - Missing or unoperative parts improperly assembled.
- i - Dimensions-incorrect dimensions of shaft and mounting.

2 - Minor
- a - Insulation damaged or deteriorated, but short circuit not possible.
- b - Scratches, cuts, abrasions, small amount of flaking, peeling or chipping.
- c - Tight wiring with no possibility of breakage.
- d - Useless holes or slots.
- e - Dimensions-incorrect overall dimensions.

4.2.2 Terminal or lead strength. -

4.2.2.1 Pull. - A pull of 5 pounds unless otherwise specified shall be applied to each lead or terminal at the point where the lead from the external circuit connects to it. Where a contacting device (e.g., cable connector, coaxial connector, or similar device) is used, the pull shall be applied to the assembled unit. The pull may be applied in any direction, and shall be increased gradually to the specified magnitude and held at that value for at least 30 seconds. One direction of pull shall be that tending to rotate any terminal.

4.2.2.2 Bend. - Any terminal that shows visible bending in the pull test (see 4.2.2.1) shall be bent back and forth five times to an angle of 45° each side of center (total 90°).

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4.2.2.3 Twist. - Stud type terminals shall be subjected to the torque listed below:

Screw Size	Torque (pound-inches)
4-40	6.5
6-40	7.6
6-32	9.0
8-36	11.5
8-32	13.5
10-32	16.0
10-42	17.5
1/4-28	30.0
1/4-20	33.0

4.2.3 Insulation resistance. - Measurements shall be made at 500 D.C. volts. The test voltage shall be applied between ground (frame) and each winding which is not permanently grounded by soldering, brazing, or welding, with all other windings grounded. Windings which are normally connected together in service are considered as one winding for the purpose of this test. (See 3.11).

4.2.4 Dielectric strength. - The test voltage shall be of commercial-line frequency and of a magnitude of 900 volts rms except for motors 1/2 horsepower and larger of motors having rated voltage greater than 250 volts, in which case a test voltage of twice rated voltage plus 1,000 volts shall be applied. The test voltage shall be applied for 1 minute, except that for inspection tests, 120 percent of the test voltage may be applied for 1 second. The test voltage shall be applied between ground(frame) and winding which is not permanently grounded by soldering, brazing, or welding, with all other windings grounded. Windings which are normally connected together in service are considered as one winding for the purpose of this test. (See 3.12).

4.2.5 Concentricity. - The motor shall be rotated slowly and examined for rubbing and binding. (See 3.13).

4.2.6 Dynamic balance. - The completely assembled motor shall be elastically mounted so that the mounting is compressed by the weight of the motor to at least the amount shown below:

Synchronous speed (rpm)	Minimum compression (inch)
900	1
1,200	9/16
1,800	1/4
3,600	1/16
7,200	1/64

The static deflection of the mounting shall in no case be greater than 50 percent of the distance between the top of the mounting in its unloaded position and the top of the mounting in its totally compresses state. With the motor mounted and operated at

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rated voltage and no load, the vibration of the bearing housing shall be measured with a vibration indicator, in the direction giving the maximum amplitude. (See 3.14).

4.2.7 Speed-torque characteristics. -

4.2.7.1 Locked rotor. - The locked-rotor test shall be performed at rated frequency and rated voltage using, preferably, the cord-and-pulley method to measure minimum locked-rotor torque and maximum locked-rotor current. Measurements shall be made using both directions of rotation if a reversible motor is specified. Measurements made during the locked-rotor test should be made as quickly as possible to avoid overheating of the motor. (See 3.26).

4.2.7.2 Full load. - The full-load test shall be performed at maximum ambient temperature during preproduction tests but may be performed at standard test conditions (see 4.1.2) during inspection tests (see 3.26).

4.2.7.2.1 Step 1. - Full-load torque and rated voltage (see 3.26) shall be applied to the motor. Before the motor temperature has increased appreciably, the full-load speed (cold) shall be measured.

4.2.7.2.2 Step 2. - The motor shall be operated at full-load torque (see 3.26) until temperature equilibrium is reached, after which measurements of full-load current and full-load speed shall be made.

4.2.7.3 Efficiency. - The efficiency test shall be performed at maximum ambient temperature. Full-load torque, rated frequency, and rated voltage shall be supplied to the motor. After the motor has reached thermal equilibrium, the power input shall be measured. (See 3.26). Efficiency shall be computed by use of the following formula:

$$\text{Efficiency} = \frac{\text{Torque (oz-in., oz-ft., or ft-lb x speed (rpm)}}{\text{K x power input (watts)}}$$

Where: K = 7.04 when the torque is given in ft-lb.

K = 112.7 when the torque is given in oz-ft.

K = 1,352 when the torque is given in oz-in.

4.2.7.4 No load. - The no-load test shall be performed at maximum ambient temperature during preproduction tests but may be performed at standard test conditions (see 4.1.2) during inspection tests. Rated voltage at rated frequency shall be applied to the motor with no load. No-load speed and no-load current shall be measured. (See 3.26).

4.2.7.5 Breakdown. - The breakdown test shall be performed at rated frequency and rated voltage. The minimum breakdown torque shall be determined by any suitable means satisfactory to the Government. (See 3.26).

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4.2.7.6 Other speed-torque characteristics. - The following speed-torque characteristics shall be determined, when applicable. The rated frequency and rated voltage shall be as specified. (See 3.26).

(a) Pull-up torque.

(b) Switching torque.

(c) Switching speed.

4.2.8 Temperature rise. - This test shall be performed at maximum ambient temperature during preproduction tests but may be performed at standard test conditions (see 4.1.2) during inspection tests. (See 3.26).

4.2.8.1 Continuous motors. - The temperature rise of the main winding shall be determined by resistance measurements in an area free of drafts, with the motor mounted by normal means on a surface of low thermal conductivity such as wood. The motor shall be operated at full-load torque until the temperature of the frame becomes stabilized. The temperature of the frame, measured on the external surface at approximately the center of the stator-winding slots, shall be determined by a thermocouple or thermometer. The resistance of the main winding shall be measured at 30-second intervals, starting immediately after the motor has been stopped, and shall be extrapolated back to the resistance at the time of shutdown for determination of R. The temperature rise of the windings shall be computed by the following formula:

$$\Delta T = \frac{R - r}{r} (t + 234.5) - (T - t)$$

Where: ΔT = temperature rise, in degrees centigrade.

R = hot resistance at time of shutdown, in ohms.

r = cold resistance, in ohms.

t = temperature of motor at time of measurement of r, in degrees centigrade.

T = mean ambient temperature during final quarter of test, in degrees centigrade.

The ambient temperature shall not vary by more than 2°C during the final quarter of this test.

4.2.8.2 Intermittent motors. - The temperature-rise test for intermittent motors shall be performed in the same manner as for continuous motors except that the applicable duty cycle shall be employed. The temperature of the frame shall be considered stabilized when the temperature peaks caused by the "on" portions of the cycle reach a maximum value which is repeated for a series of consecutive cycles.

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4.2.9 Temperature cycling. (See 3.15). - Motors shall be subjected to the temperature cycling test specified in method 102 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition letter - A.

(b) Initial measurements - Not applicable.

(c) Final measurements - During the last cycle, immediately after removal from the test chamber following step 1, the motor shall be subjected to the dielectric-strength test for a period of 5 seconds at 90 percent of the test voltage specified in 4.2.4.

4.2.10 Vibration (see 3.16). - Motors shall be subjected to the vibration test specified in method 201 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - By normal mounting means, except that shock mounts shall not be used unless they are an integral part of the motor.

(b) Initial measurements - Not applicable.

(c) Motion - The motor shall be vibrated for a total of 7-1/2 hours, 2-1/2 hours in each of three mutually perpendicular directions. During the first 1-1/4 hours of vibration in each of these directions, the motor shall not be operated. During the remaining period of vibration in each of these directions, the motor shall be operated in accordance with the applicable duty cycle at rated voltage and at no-load.

(d) Final measurements - Not applicable.

4.2.11 Mechanical shock. - The motor shall be tested in accordance with test I or II, as specified.

4.2.11.1 Shock test 1. - The motor shall be operated at rated voltage and no load during the mechanical shock test.

4.2.11.1.1 Equipment and mounting. - The shock-testing machine shall be that specified for lightweight equipment in Specification MIL-S-901. The motor shall be firmly secured, in its normal mounting position, to the mounting plate of the shock-testing machine. All shock-absorbing devices used on the motor shall be in place and unblocked. No shock mounts shall be used unless they are an integral part of the motor.

4.2.11.1.2 Number of blows. - Three blows shall be applied parallel to each of the three principal axis of the motor for a total of nine blows. The height of the hammer drop for each axial blow and each side blow shall be successively 1 foot, 2 feet, and 3 feet. The height of the hammer drop for each top blow shall be successively 2 feet, 3 feet, and 4 feet. (See 3.18).

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4.2.11.2 Shock test II. - The motor shall be tested in accordance with Specification 7201. An acceleration of 30 gravitational units shall be applied in 10 milliseconds.

4.2.12 Voltage and frequency variation (see 3.19). - The motor shall be operated for 2 hours at full-load torque in accordance with the applicable duty cycle, under each of the following conditions:

(a) At 110 percent rated voltage and 105 percent rated frequency.

(b) At 110 percent rated voltage and 95 percent rated frequency.

(c) At 90 percent rated voltage and 105 percent rated frequency.

(d) At 90 percent rated voltage and 95 percent rated frequency.

4.2.13 Overload. - The overload test shall be performed at maximum ambient temperature for preproduction tests but may be performed at standard test conditions (see 4.1.2) for inspection tests. The motor shall be operated at rated voltage with 125 percent of full-load torque (see 3.26) applied for a period of 2 hours. After this test, the no-load speed shall be measured and the dielectric-strength test shall be performed at 90 percent of the test voltage specified in 4.2.4. (See 3.17).

4.2.14 Low-temperature storage and operation (when applicable). - The motor shall be placed within the chamber and cooled to and maintained at a temperature of $-65^{\circ}C\ {}^{+0}_{-3^{\circ}C}$ for a period of 48 hours. The temperature of the chamber shall then be raised to $-55^{\circ}C\ {}^{+0}_{-3}$ and maintained for 24 hours. At the conclusion of this exposure period and while at this temperature, the motor shall be started with full-load torque applied at rated voltage, and the time needed to reach 90 percent of full-load speed shall be measured. Continuous motors shall operate at full-load torque for 1/2 hour, and intermittent motors shall operate at the applicable duty cycle for 1/2 hour of elapsed time or six cycles, whichever is longer.

4.2.15 Moisture resistance. (See 3.21). - Motors shall be subjected to the moisture-resistance test specified in Method 106 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - On racks by normal mounting means.

(b) Initial measurements - Not applicable.

(c) Polarization and Load - Not applicable.

(d) Measurements. - At the completion of the test and within one hour after removal of the motor from the chamber, the insulation resistance shall be measured. The motor shall then be allowed to dry from 4 to 6 hours at a temperature of $25^{\circ}C \pm 5^{\circ}C$ and a relative humidity of 40 to 60 percent in order to remove surface moisture, after which the dielectric-strength test at 90 percent of the test voltage

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specified in 4.2.4 shall be performed. The motor shall then be operated for 1 hour at full-load torque, after which the full-load speed shall be measured. The insulation resistance shall again be measured after the motor has been allowed to dry for at least 48 hours at a temperature of 25°C ± 5°C and a relative humidity of 40 to 60 percent. (See 3.21.)

4.2.16 Life (See 3.26). - During the life test, precautions shall be taken against undue air contamination of a type not to be expected in normal service use. Reversible motors shall be operated in both directions for an approximately equal number of hours.

4.2.16.1 Continuous motors. - The life test shall be performed for a total of 2500 elapsed hours unless otherwise specified on Complementary Documents. The motor shall be operated in normal mounting position at rated voltage, full-load torque, and maximum ambient temperature. During the first 100 elapsed hours, a switching cycle of 2 minutes on and 1 minute off shall be employed. For the remaining 900 hours, the motor shall be operated continuously. After the life test, the motor shall be subjected to the dielectric-strength test at 75 percent of the test voltage specified in 4.2.7 to the dynamic balance (see 3.14) and the full-load speed and full-load current shall be measured. These measurements may also be made at any time during the test when the proper performance of the motor is questionable.

4.2.16.2 Intermittent motors. - The life test on intermittent motors shall be performed for a total of 1000 hours unless otherwise specified with the motor operating in accordance with the applicable duty cycle and at maximum ambient temperature. Measurements shall be made as specified for continuous motors.

4.2.17 Corrosion (when applicable). - Motors shall be subjected to the salt-spray (corrosion) test specified in method 101 of Standard MIL-STD-202. The following details shall apply:

(a) Test condition letter - B.

(b) Standards of acceptability - Upon completion of the test, the motor shall be subjected to the full-load test (see 4.2.7.2) standard test conditions (see 4.1.2), and the markings shall be examined for legibility.

4.2.18 Motor enclosure. -

4.2.18.1 Explosionproof. - The motor shall be tested in accordance with explosion-proof (aeronautical) test procedure II of Specification MIL-E-5272. (See 3.23.1).

4.2.18.2 Dustproof. - The motor shall be tested in accordance with sand-and-dust test procedure II of Specification MIL-E-5272. While in the test chamber the motor shall be operated for 30 minutes at no load and shall then be at rest for 90 minutes. This cycle shall be repeated four times (a total of 8 hours). After this test the motor shall be subjected to step 2 of the full load test (see 4.2.7.2.2) at the standard test conditions specified in 4.1.2 (See 3.23.10).

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4.2.18.3 Oilproof. - During this test the motor in its normal mounting position shall be operated at rated voltage with no load, at the applicable duty cycle. Hydraulic oil conforming to Specification MIL-O-5606, maintained at 69° ± 5°C., shall be dripped on the motor at a rate of at least 1/2 pint per hour and allowed to flow over the motor. This test shall be performed for a period of 48 hours. After this test the motor shall be cleaned and dried and then subjected to the dielectric-strength at 50 percent of the test voltage specified in 4.2.4. (See 3.23.9).

4.2.18.4 Watertight. - The motor shall be mounted in the normal position and subjected to a stream of water from a 1-inch nozzle with a velocity head of 35 feet, held at a distance of 10 feet from the motor. The stream shall be applied from each direction which is likely to cause water penetration. The duration of the test shall be 60 minutes, consisting of 20 minutes with the motor at rest, 20 minutes with the motor operating at no load, and the final 20 minutes with the motor at rest. After this test the motor shall be disassembled sufficiently to permit inspection for water penetration. (See 3.23.2).

4.2.18.5 Submersible. - The motor shall be submerged in sea water having a specific gravity of 1.025 under a pressure equivalent of a 50-foot head for a period of 1 hour. After this test the motor shall be disassembled sufficiently to permit inspection for water penetration. (See 3.23.3).

4.2.19 Tropicalization. - The motor shall be tested in accordance with fungus resistance test Procedure I of Specification MIL-E-5272. At the conclusion of the test, the no-load speed shall be measured at standard test conditions (see 4.1.2) and the dielectric-strength test shall be performed at 75 percent of the test voltage specified in 4.2.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Domestic, immediate use. - Motors, Alternating Current, Fractional Horsepower (Squirrel Cage; For Special Military Applications) shall be preserved and packaged to afford protection against deterioration and damage during shipment.

5.1.2 Domestic limited storage. - Motors, Alternating Current, Fractional Horsepower (Squirrel Cage; For Special Military Applications) shall be packaged Method III, per Specification MIL-P-116 as follows: Cushion each motor by wrapping in fiberboard, corrugated, single faced (Flexible) conforming to Specification LLL-F-291. Place each cushioned motor up to 5 pounds in weight within a close fitting setup box, complying with Specification JAN-P-133. Seal entire closure with Tape, Paper, Gummed, Sealing and securing conforming to Specification UU-T-111.

5.1.2.1 Motors over 5 pounds in weight. - Motors over 5 pounds in weight shall be packaged Method III, using the applicable procedures as specified in 5.1.2, except each cushioned motor shall be placed within a fiber corrugated box conforming to Specification LLL-B-631, seal entire closure with Tape, Paper, Gummed conforming to Specification UU-T-111.

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5.1.3 Military overseas. - Preserve and package Motors, Alternating Current, Fractional Horsepower (Squirrel Cage; For Special Military Applications) in accordance with procedures specified for designated methods as prescribed in Specification MIL-P-116 and further described in the following paragraphs.

5.1.3.1 Exterior shafts, couplings, and all unpainted exposed metal surfaces of all motors shall be cleaned and preserved prior to placement within the unit package following procedures as specified in MIL-P-116 for C-1 cleaning and P-7 or P-10 preservatives. Wrap the preserved metal parts in greaseproof barrier material, grade A, conforming to Specification JAN-B-121, of the lightest practicable weight type to resist penetration of cutting edges, and secure by suitable tape.

5.1.3.2 Motors (weights up to 10 pounds). - Motors up to 10 pounds in weight shall be packaged individually Method IId, as follows: Cushion each motor by wrapping in fiberboard, corrugated, single faced (flexible conforming to Specification LLL-F-291.) Place each cushioned motor within a metal container Type III provided with a key wind, rip strip opening device. Conforming to Specification MIL-C-124.

5.1.3.3 Motors (weights fractional 10 pounds up to 1 horsepower 90 pounds). - Package each fractional motor individually weighing 10 pounds to 90 pounds Method IIb, as follows: Cushion each motor on all surfaces with cells or pads or both fabricated of corrugated fiberboard, conforming to Specification 19-10 (A or C flute acceptable) and designed to absorb the shock of impact normally encountered in handling and transit. Place each cushioned motor together with the required amount of desiccant as prescribed in Specification MIL-P-116, within a close-fitting suitable style and grade corrugated fiberboard box. Seal entire closure with gummed Kraft tape conforming to Specification UU-T-111. Blunt all corners of the box. Place the boxed motor within a water-vapor-proof barrier, qualified for Class 2 up to 25 pounds and class 1 for 25 to 90 pounds application in accordance with the requirements of Specification MIL-B-131. Heat seal closure. Place the water-vapor-proofed motor within a close fitting suitable style and grade fiberboard box conforming to Specification JAN-P-108. Seal entire box closure, corners and joints with water-resistant pressure-sensitive tape, conforming to Specification JAN-P-127.

5.2 Packing. -

5.2.1 Domestic, immediate use. - Motors, Alternating Current, Fractional Horsepower (Squirrel Cage; For Special Military Applications) shall be so packed as to insure arrival at destination in satisfactory condition and in containers complying with Consolidated Freight Classification Rules at lowest rates.

5.2.2 Domestic, limited storage. - Motors, Alternating Current, Fractional Horsepower (Squirrel Cage; For Special Military Applications) shall be packed in wood cleated fiberboard, wood cleated plywood, nailed wood, or wirebound boxes, conforming to Specifications PPP-B-601, NN-B-621, or NN-B-631. The gross weight shall not exceed 200 pounds for plywood or wood boxes. Closure shall be made in accordance with the applicable box specification.

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5.2.3 Military,overseas.- The Motors, Alternating Current, Fractional Horsepower, (Squirrel Cage; For Special Military Applications) shall be packed in a wood cleated plywood, or nailed wood box conforming to the requirements of Specifications PPP-B-601 or JAN-P-106, respectively. The approximate gross weight of any exterior shipping container shall not exceed 150 pounds. Close the shipping containers, as specified in the appendix of the applicable box specification.

5.2.3.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.3 Marking for shipment.- Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of MIL-STD-129.

6. NOTES

6.1 Ordering data.-

6.1.1 Motors for which specification sheets have been issued.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Title, number, and date of the applicable specification sheet. (see 3.26)

(c) Whether packing and marking are for domestic or overseas shipment. (See Section 5)

6.1.2 Motors for which specification sheets have not been issued.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) The information required on figure 1.

(c) Whether packing and marking are for domestic or overseas shipment. (See Section 5)

6.1.3 Indirect shipment.- The packaging, packing, and marking specified in section 5 apply only to direct purchases by or direct shipments to the Government and are not intended to apply to contracts or orders between the manufacturer and prime contractor.

6.2 Design and application of motors.- Matters relating to the design and application of motors covered by this Specification should be referred to the technical service whose equipment is involved.

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6.3 Data need to establish specification sheets. - The information listed in item 6.1.2(b) will be the basis for establishing an individual specification sheet. Any additional information (i.e., test data) should also be included with the request for establishment of a specification sheet.

6.4 Sampling procedure. - The sampling procedure specified for group B inspection tests (see 4.1.5.2) is identified for reference purposes as plan B-2.

6.5 Speed-torque curve. - Figure 2 shows a typical dimensionless speed-torque curve for motors covered by this specification. By plotting the speed-torque values contained in the applicable individual specification sheet, etc., the curve for a particular motor may be obtained.

6.6 Ratings. - In order to accomplish standardization of motors for Military applications, motors should conform to the following ratings, whenever possible:

Voltage Ratings

115 volts (1- or 3- phase)
208 volts (3-phase)
230 volts (1-phase)

TABLE VII - Full-load-speed ratings. 1/

60 cps	50 cps	25 cps
rpm	rpm	rpm
3,450	2,850	1,425
1,725	1,425	
1,140	950	
850		

1/ These ratings are not applicable to shaded-pole and permanent-split-phase capacitor motors.

TABLE VIII - Maximum ambient temperatures and maximum temperature rises (continuous motors only) 1/

Class of insulation 1/	Maximum ambient temperature	Maximum temperature rise	
		Open or protected motor	Other enclosure
	°C	°C	°C
A-----	40	60	65
A-----	50	50	55
B-----	40	80	85
B-----	50	70	75
B-----	85	35	40

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1/ Class H insulation is generally used in particular applications and should be specified on an individual basis.

TABLE IX - Duty Cycles.

Motor Designation	Full Load	Rest
Intermittent A	5 minutes	15 minutes
Intermittent B	1 minute	29 minutes

6.7 Basis of horsepower ratings. - The horsepower ratings assigned to motors are determined by breakdown torque, in accordance with table X.

6.8 Insulation materials. -

6.8.1 Class A. - Class A insulation consists of:

(a) Cotton, silk, paper, and similar organic materials, either impregnated with or immersed in a liquid dielectric.

(b) Molded or laminated materials with cellulose filler, phenolic resin, or other resins of similar properties.

(c) Films and sheets of cellulose acetate and other cellulose derivatives of similar properties.

(d) Varnish (enamel) as applied to conductors.

6.8.2 Class B. - Class B insulation consists of mica, asbestos, fiber glass, and similar inorganic materials in built-up form with organic binding substances. A small portion of Class A materials may be used for structural purposes only. The electrical and mechanical properties of the insulated winding should not be impaired by application of the temperatures for class B materials.

6.8.3 Class H. - Class H insulation consists of:

(a) Mica, asbestos, fiber glass, and similar inorganic materials in built-up form with binding substances composed of silicone compounds or materials with equivalent properties.

(b) Silicone compounds in rubbery or resinous forms or materials with equivalent properties.

A minute proportion of class A materials may be used in class H materials only where essential for structural purposes during manufacture.

6.9 Frame sizes. - In order to accomplish standardization of motors for military applications, motors should conform to the frame sizes indicated in Figure 3 wherever possible.

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IDENTIFICATION FORM FOR MOTOR,
ALTERNATING CURRENT, FRACTIONAL HORSEPOWER
(SQUIRREL CAGE: FOR SPECIAL MILITARY APPLICATIONS)

1. Motor identification:_____.

2. Motor type (such as capacitor-run, split- phase, etc.):_____.

3. Horsepower rating (nominal):_____horsepower.

4. Duty cycle:_____.

5. Temperature characteristics:
   Maximum ambient temperature:_____°C.
   Maximum temperature rise of winding_____°C.
   Maximum temperature rise of frame:_____°C.

6. Speed-torque characteristics (as applicable):
   Capacitor rating:_____microfarads,_____volts, working voltage
   Rated frequency:_____cycles per second
   Rated voltage:_____volts
   Full-load torque:_____ounce-foot or ounce- inch
   Type of loading (such as fan load, dynamometer, etc.):_____.
   Locked-rotor torque:_____ounce-foot or ounce-inch minimum.
   Locked-rotor current:_____amperes, maximum
   Full-load speed:_____rpm± _____rpm or percent.
   Full-load current:_____ amperes, maximum
   Full-load power input:_____watts, maximum
   Efficiency at full load:_____percent, minimum
   No-load speed:_____rpm ± _____rpm or percent
   No-load current:_____amperes maximum
   Breakdown torque:_____ounce-foot or ounce-inch, minimum
   Pull-up torque:_____ounce-foot or ounce-inch, minimum
   Switching torque:_____ounce-foot or ounce-inch, minimum
   Switching speed:_____rpm ± _____rpm or percent
   Power Factor (nominal):_____

7. Direction of rotation:_____.

8. Type of enclosure:_____.

9. Relubrication required during expected life (yes or no):_____.
   Relubricate after:_____hours.

Figure 1. Motor-Identification Form

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10. Type of bearings (sealed ball bearings, ball bearings, sleeve bearings, etc.); _____

11. Type and length of electrical connectors: _____.
Pull-test requirement (when applicable): _____pounds.

12. End-play requirements:
Spring-loaded (yes or no): _____.
Maximum free end-play: _____inches.

13. Maximum weight: _____pounds.

14. Motor-operating-life expectancy: _____years or hours.
Life test required _____hours at _____degrees centrigrade.

15. Overload protection (yes or no): _____.

16. Operating-position limitations (horizontal only, vertical only, no limitations, etc.,): _____.

17. Low-temperature storage and operation (applicable or not applicable): _____

18. Special features and requirements (such as steel bearing inserts, etc): _____
_____
_____

19. Circuit diagram, showing color code:

20. Low-temperature starting time

21. Low-temperature operating time

22. Mechanical-shock test (Test 1 or 11)

23. Corrosion test (applicable or not applicable)

24. Class of insulation

25. Frame, mounting, electrical-connector, and mechanical-coupling diagram (sketch, two views), showing envelope, mounting, and mechanical-coupling dimensions and tolerances. (Show only details which will affect interchangeability.)

Figure 1. Motor-Identification Form (Contd).

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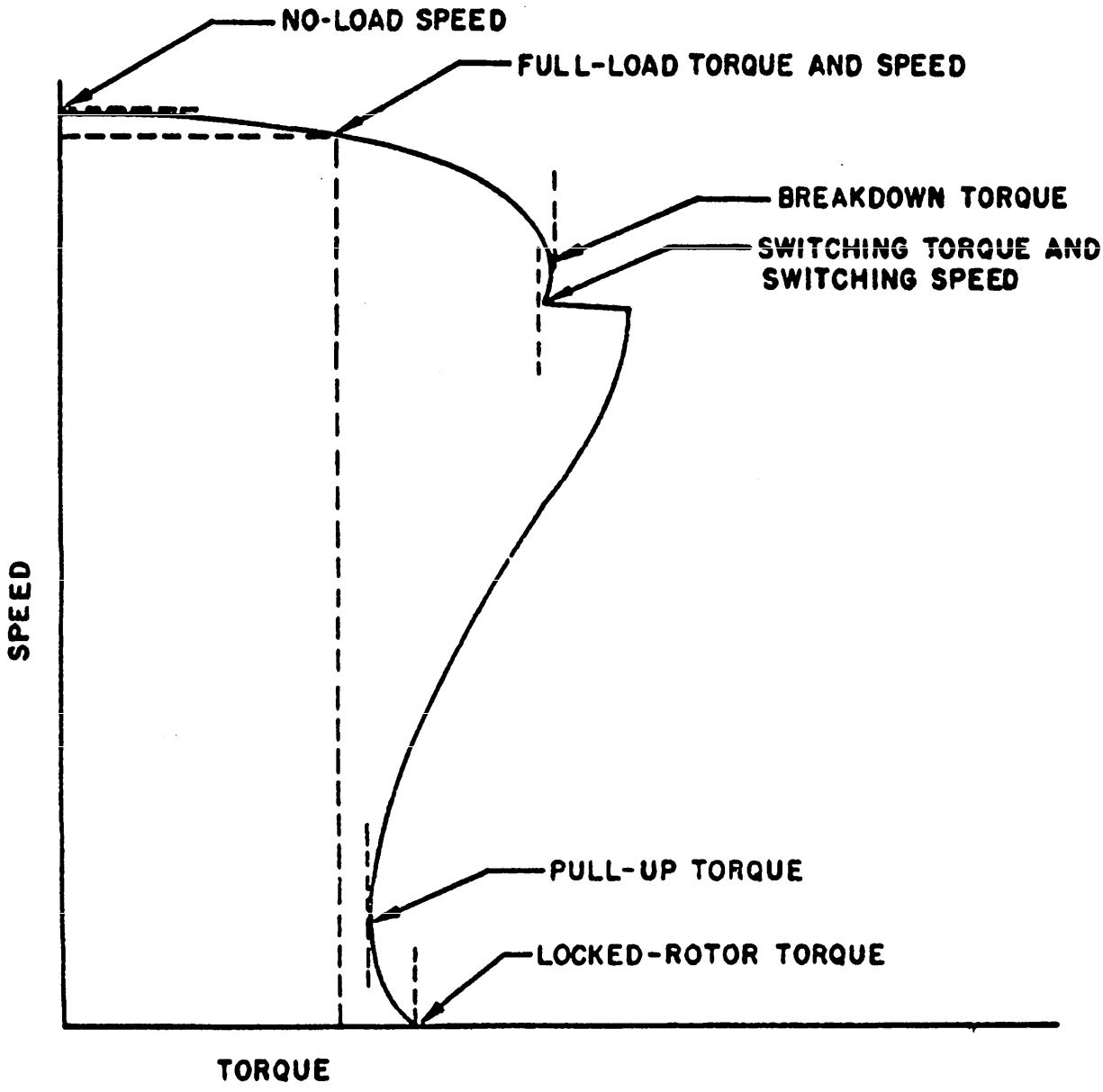
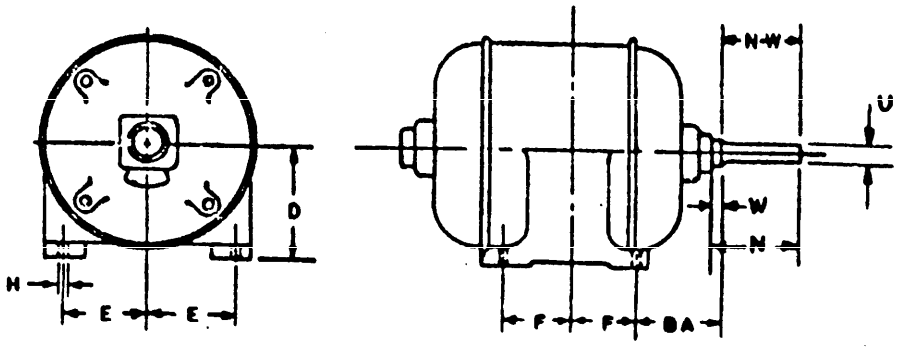


FIGURE 2. SPEED_TORQUE CURVE OF A SPLIT-PHASE MOTOR

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Frame No.	D	E	F	BA	H	N-W*	U
42	2-5/8**	1-3/4	27/32	2-1/16	9/32 slot	1-1/8	3/8
48	3**	2-1/8	1-3/8	2-1/2	9/32 "	1-1/2	1/2
56	3-1/2**	2-7/16	1-1/2	2-3/4	11/32 "	1-7/8	5/8
66	4-1/8**	2-15/16	2-1/2	3-1/8	13/32 "	2-1/4	3./4

Frame No.	Key		
	Wide	Thick	Long
42	3/64	Flat	
48	3/64	Flat	
56	3/16	3/16	1-3/8***
66	3/16	3/16	1-7/8***

*N-W is the length of shaft extensio n from shoulder to shaft end.
** Dimension D will never be greater than the above values for rigid-base motors., but it may be less so that shims are usually required for coupled or geared machines.
*** Effective length of keyway.

STANDARD DIMENSIONS FOR NEMA FHP FRAME SIZES

(Form NEMA Standard MG1-3. 2)

Figure 3

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Table X - Horsepower Ratings 1/

Horsepower	Synchronous Speed						
	3,600 rpm	3,000 rpm	1,800 rpm	1,500 rpm	1,200 rpm	1,000 rpm	900 rpm
	Breakdown Torque (ounce-inches) 2/						
0.001	0.35 - 0.55	0.42 - 0.66	0.7 - 1.1	0.85 - 1.3	1.1 - 1.65		
.0015	.55- .7	.66 - .85	1.1 - 1.45	1.3 - 1.75	1.65 -2.2		
.002	..7 - 1.1	.85 - 1.3	1.45- 2.2	1.75 - 2.6	2.2 - 3.3		
.003	1.1 - 1.8	1.3 - 2.2	2.2 - 3.6	2.6 - 4.3	3.3 - 5.4		
.005	1.8 - 2.7	2.2 - 3.2	3.6 - 5.4	4.3 - 6.6	5.4 - 8.1		
.0075	2.7 - 3.6	3.2 - 4.3	5.4 - 7.2	6.6 - 8.6	8.1 -11		
.010	3.6 - 5.5	4.3 - 6.6	7.2 - 11	8.6 - 13	11 -17		
.015	5.5 - 9.5	6.6 -11.4	11 - 19	13 - 23	17 -29		
.025	9.5 -15	11.4 -18	19 -30	23 - 36	29 -46		
.035	15 -24	18 -28.8	30 -48	36 - 57.6	46 -72		
	Breakdown Torque (ounce-feet) 2/						
1/20	2.0 - 3.6	2.4 - 4.4	4.0 - 7.1	4.8 - 8.5	6.0 - 10.4	7.2 - 12.4	8.0 - 13.5
1/12	3.7 - 6.0	4.4 - 7.2	4.1 - 11.5	8.5 - 13.8	10.4 - 16.5	12.4 - 19.8	13.5 - 21.5
1/8	6.0 - 8.7	7.2 - 10.5	11.5 - 16.5	13.8 - 19.8	16.5 - 24.1	19.8 - 28.9	21.5 - 31.5
1/6	8.7 - 11.5	10.5 - 13.8	16.5 - 21.5	19.8 - 25.8	24.1 - 31.5	28.9 - 37.8	31.5 - 40.5
1/4	11.5 - 16.5	13.8 - 19.8	21.5 - 31.5	25.8 - 37.8	31.5 - 44.0	37.8 - 53.0	40.5 - 58.0
1/3	16.5 - 21.5	19.8 - 25.8	31.5 - 40.5	37.8 - 48.5	44.0 - 58.0	53.0 - 69.5	58.0 - 77.0
1/2	21.5 - 31.5	25.8 - 37.8	40.5 - 58.0	48.5 - 69.5	58.0 - 82.5	69.5 - 99.0	
3/4	31.5 - 44.0	37.8 - 53.0	58.0 - 82.5	69.5 - 99.0			
1	44.0 - 58.0	53.0 - 69.5					

1/ Horsepower ratings for 400-cps motors are inversely proportional to synchronous speed.

2/ The breakdown-torque range includes the maximum torque, but does not include the minimum torque.

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6.10 Contractual requirements. - It is recommended that the following be included in the contract:

6.10.1 Additional inspection. - It should be understood that additional examination, measurement, and nondestructive testing of equipment may be performed by the Government when considered necessary to determine compliance with this specification and other applicable documents. Accordingly, the Government inspector may withdraw materials, parts, subassemblies, or components temporarily from production for such inspection performed either at a Government laboratory or the contractor's plant.

6.10.2 Waiver of preproduction samples. - When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should be made by the prime contractor and should include a statement indicating that: the specified samples have been submitted amd approved on the current contract: production of the item under the new order will be concurrent with, or follow without interruption, production on the current order and will be at the same plant; and current production is in strict compliance with specified requirements for the new equipment except for such deviations as may be authorized on the current order. These deviations should be described in the statement. Waiver of preproduction samples may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8378

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